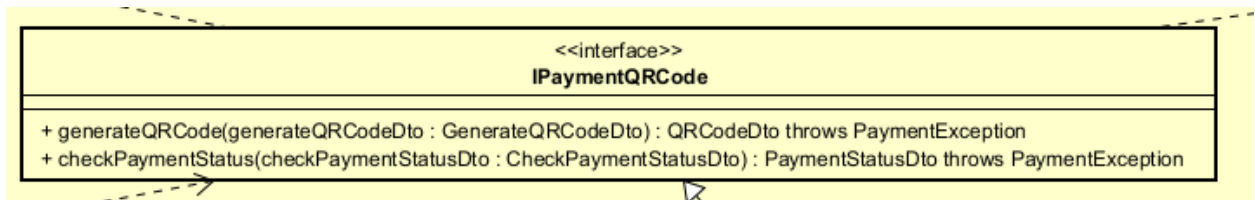


Design for interface "IPaymentQRCode"



Description: An interface defining the standard contract for QR code-based payment processing. It ensures a consistent implementation across different payment gateways (like VietQR) for generating payment QR codes and verifying transaction statuses.

Table 1. Operation design

#	Name	Return type	Description (purpose)
1	generateQRCode	QRCodeDto	Define the contract for generating a payment QR code based on order details.
2	checkPaymentStatus	PaymentStatusDto	Define the contract for querying and verifying the status of a specific QR payment transaction.

Detailed Operation Design

1. Operation: generateQRCode

Parameter:

- generateQRCodeDto: GenerateQRCodeDto The data transfer object containing the necessary order and payment details (such as amount, content, and account info) required to generate a QR code.

Exception:

- PaymentException if the QR generation fails due to invalid input data, missing required fields, or errors from the underlying payment provider's API.

Method:

- Receive the generateQRCodeDto containing the payment request details.

- Process the data to interface with the specific payment gateway implementation (e.g., VietQR) to create the QR code.
- Return a QRCodeDto containing the generated QR code string or image link for the client to display.

2. Operation: checkPaymentStatus

Parameter:

- checkPaymentStatusDto: CheckPaymentStatusDto The data transfer object containing transaction identifiers (like transaction ID or order reference) needed to look up the payment status.

Exception:

- PaymentException if the status check fails due to missing identifiers, network errors, or if the underlying payment provider's API encounters an error.

Method:

- Receive the checkPaymentStatusDto to identify the specific transaction.
- Query the underlying payment gateway or internal system to retrieve the most up-to-date state of the transaction.
- Map the result and return a PaymentStatusDto representing the current status of the payment.

Design for class “VietqrService”

VietqrService	
- clientId : string - apiKey : string - accessToken : string - baseUrl : string	
+ generateQRCode(generateQRCodeDto : GenerateQRCodeDto) : QRCodeDto throws PaymentException + checkPaymentStatus(checkPaymentStatusDto : CheckPaymentStatusDto) : PaymentStatusDto throws PaymentException + handleCallback(vietqrCallbackDto : VietqrCallbackDto) : PaymentStatusDto throws PaymentException + buildGenerateRequest(generateQRCodeDto : GenerateQRCodeDto) : VietqrGenerateRequestDto + mapGenerateResponse(response : VietqrGenerateResponseDto) : QRCodeDto + mapCallbackToPaymentStatus(vietqrCallbackDto : VietqrCallbackDto) : PaymentStatusDto + mapTransactionStatusResponse(response : VietqrTransactionStatusResponseDto) : PaymentStatusDto	

Table 1. Example of attribute design

#	Name	Datatype	Default value	Description
1	clientId	string		Client ID used to call the partner API
2	apiKey	string		API key for authentication
3	accessToken	string		Access token (typically used after fetching the token)
4	baseUrl	string		Base URL of the VietQR API system

Table 2. Example of operation design

Operation 1

Bảng

#	Name	Return type	Description (purpose)
1	generateQRCode(generateQRCodeDto : GenerateQRCodeDto)	QRCodeDto	Generate QR code for payment transactions

#	Name	Return type	Description (purpose)
2	checkPaymentStatus(checkPaymentStatusDto : CheckPaymentStatusDto)	PaymentStatusDto	Check payment status of the transaction
3	handleCallback(vietqrCallbackDto : VietqrCallbackDto)	PaymentStatusDto	Process callback data from VietQR
4	buildGenerateRequest(generateQRCodeDto : GenerateQRCodeDto)	VietqrGenerateRequestDto	Build request payload to call the QR code generation API
5	mapGenerateResponse(response : VietqrGenerateResponseDto)	QRCodeDto	Map/normalize the generation response from VietQR
6	mapCallbackToPaymentStatus(vietqrCallbackDto : VietqrCallbackDto)	PaymentStatusDto	Map callback DTO to PaymentStatusDto
7	mapTransactionStatusResponse(response : VietqrTransactionStatusResponseDto)	PaymentStatusDto	Map transaction status response to PaymentStatusDto

Detail Operation

1. Operation: generateQRCode(generateQRCodeDto: GenerateQRCodeDto): QRCodeDto throws PaymentException

Parameter:

- generateQRCodeDto: GenerateQRCodeDto
- + amount

- + content
- + accountNo
- + accountName
- + bankCode

Exception:

- `PaymentException` (Failed to generate VietQR code due to VietQR API calling error, token authentication error, or invalid request data).

Method:

- Receive `generateQRCodeDto` from the client to create the VietQR generation request information.
- Use `buildGenerateRequest(generateQRCodeDto)` to convert the data into `VietqrGenerateRequestDto`.
- Call `VietqrApiClient.getAccessToken()` to retrieve the access token (if not available or expired).
- Call `VietqrApiClient.generateQRCodeRequest(vietqrGenerateRequestDto, accessToken)` to generate the QR code on the VietQR system.
- Use `mapGenerateResponse(response)` to map the VietQR response to internal data.
- Return `QRCodeDto` to the controller/client layer (including QR information for payment: QR code/link and related transaction details).

2. Operation: `checkPaymentStatus(checkPaymentStatusDto: CheckPaymentStatusDto): PaymentStatusDto` throws `PaymentException`

Parameter:

- `checkPaymentStatusDto: CheckPaymentStatusDto`
 - + `transactionId` (or the corresponding transaction identifier according to the use case).

Exception:

- `PaymentException` (Error checking VietQR transaction status due to invalid/expired token, API calling error, or non-existent transaction).

Method:

- Receive the transaction identifier information from `checkPaymentStatusDto`.
- Retrieve the `accessToken` (call `VietqrApiClient.getAccessToken()` if necessary).
- Call `VietqrApiClient.getTransactionStatus(checkPaymentStatusDto, accessToken)` to query the status on VietQR.

- Map VietqrTransactionStatusResponseDto to PaymentStatusDto using mapTransactionStatusResponse(response).
- Return PaymentStatusDto representing the transaction status so the system can update the order or respond to the client.

3. Operation: *handleCallback(vietqrCallbackDto: VietqrCallbackDto): PaymentStatusDto* throws *PaymentException*

Parameter:

- vietqrCallbackDto: VietqrCallbackDto
 - + transactionId
 - + transactionContent
 - + transactionDateTime
 - + amount
 - + status

Exception:

- PaymentException (Callback processing error due to invalid payload, data mapping error, or transaction status update/synchronization error).

Method:

- Receive callback data sent by VietQR.
- Call mapCallbackToPaymentStatus(vietqrCallbackDto) to convert the callback into PaymentStatusDto (status, transaction code, time, amount, etc.).
- Optionally call the synchronization/status re-check flow via syncTransaction(checkPaymentStatusDto) to ensure consistency with VietQR.
- Return PaymentStatusDto to the controller layer to continue business logic updates.

4. Operation: *buildGenerateRequest(generateQRCodeDto: GenerateQRCodeDto): VietqrGenerateRequestDto*

Parameter:

- generateQRCodeDto: The system's QR code generation request information (e.g., orderId, amount, currency, customerInfo/description according to your DTO design).

Exception:

- PaymentException (If the system lacks sufficient data to create a VietQR request, e.g., missing clientId/apiKey, missing orderId/amount, or invalid currency).

Method:

- Use generateQRCodeDto to generate the business fields of VietqrGenerateRequestDto according to the VietQR format (mapping to the corresponding fields).
- Attach access configuration to the request using properties of VietqrService (e.g., clientId, apiKey if the request requires them).
- Set up display content/metadata information (if the DTO requires it) to ensure the callback can be matched with the orderId.
- Return the fully built VietqrGenerateRequestDto to be used by VietqrApiClient.generateQRCodeRequest(...).

5. Operation: mapGenerateResponse(response: VietqrGenerateResponseDto): VietqrGenerateResponseDto

Parameter:

- response: The DTO returned from VietQR upon successful QR creation (e.g., qrCode, qrUrl, transactionId, orderReference, expiresAt... depending on your DTO).

Exception:

- PaymentException (If the VietQR response is missing mandatory data, e.g., missing transactionId or necessary QR/URL information).

Method:

- Read the fields from response and check for data completeness for the QR generation flow.
- Map/convert to the internal VietqrGenerateResponseDto (keeping the DTO type as required by the signature; if your DTO is an "internal standard", just standardize the fields and ensure the correct format).
- Ensure the fields needed for the next step are accurately retained (especially transactionId/orderReference to match the payment status).
- Return the fully mapped VietqrGenerateResponseDto.

6. Operation: mapCallbackToPaymentStatus(vietqrCallbackDto: VietqrCallbackDto): PaymentStatusDto

Parameter:

- vietqrCallbackDto: Callback data from VietQR (e.g., transactionId, orderReference, status, amount, timestamp, signature... depending on your DTO template).

Exception:

- PaymentException (If the callback is invalid or missing mandatory data to deduce the payment status, e.g., missing transactionId/status or orderReference).

- `PaymentException` (If the callback data cannot be authenticated or fails business validation, e.g., invalid signature — if your use case/ACD mentions callback authentication).

Method:

- Extract the status from `vietqrCallbackDto` and convert it to the corresponding enum/status value in `PaymentStatusDto`.
- Assign `orderId/transactionId` to `PaymentStatusDto` by mapping from the callback's `orderReference/transactionId`.
- Map amount/currency/time (if present in the callback) to the corresponding fields for the system to save and display to the customer.
- Return `PaymentStatusDto` so `handleCallback(...)` can update the order status in the system flow.

7. Operation: *mapTransactionStatusResponse(response: VietqrTransactionStatusResponseDto): PaymentStatusDto*

Parameter:

- `response`: The DTO returned from VietQR when querying the transaction status (e.g., `transactionId`, `status`, `amount`, `paidTime/updatedAtTime...`).

Exception:

- `PaymentException` (If the VietQR response is missing mandatory data, e.g., missing `transactionId` or `status`).

Design for class "VietqrController"

VietqrController	
+ generateQRCode(generateQRCodeDto : GenerateQRCodeDto) : QRCodeDto + handleCallback(vietqrCallbackDto : VietqrCallbackDto) : PaymentStatusDto + checkPaymentStatus(checkPaymentStatusDto : CheckPaymentStatusDto) : PaymentStatusDto	

Description: Controller class responsible for handling VietQR payment flows, including generating QR codes for orders, processing webhooks/callbacks from the VietQR system, and manually synchronizing transaction statuses.

Table 1. Operation design

#	Name	Return type	Description (purpose)
1	generateQRCode	QRCodeDto	Handle requests to generate a VietQR code for customer payment.
2	handleCallback	PaymentStatusDto	Process automated callbacks from VietQR to update transaction statuses.
3	checkPaymentStatus	PaymentStatusDto	Check the current payment status of a transaction by synchronizing directly with the VietQR gateway.

Detailed Operation Design

1. Operation : generateQRCode

Parameter:

- generateQRCodeDto: N/A, The data transfer object containing necessary order details, as VietQR is the default payment method.

Exception:

- PaymentException if failed to generate VietQR code due to API calling error, token authentication error, or invalid request data.

Method:

- Receive generateQRCodeDto from the client to create the VietQR generation request information.
- Call the VietQR API to generate the QR code using the provided integration specifications.
- Return QRCodeDto to the client, providing the QR code for the customer to scan and complete the payment.

2. Operation: handleCallback**Parameter:**

- vietqrCallbackDto: N/A, Callback data containing transaction information such as transaction ID, transaction content, and transaction datetime.

Exception:

- PaymentException if callback processing encounters an error due to invalid payload, mapping error, or transaction status update failure.

Method:

- Receive callback data to process and simulate a successful payment.
- Map the callback payload to the internal PaymentStatusDto structure.
- Ensure the software records the payment transaction information and the successfully paid order.
- Return PaymentStatusDto to update the order status within the system flow.

3. Operation: checkPaymentStatus**Parameter:**

- checkPaymentStatusDto: Contains the transaction identification information required to query and verify the payment status with the VietQR gateway.

Exception:

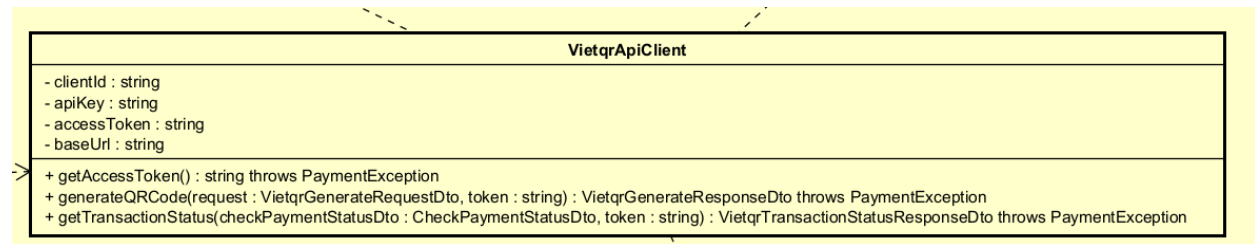
- PaymentException: Thrown when transaction synchronization fails due to API call errors, network issues, invalid gateway responses, or when the specified transaction cannot be found.

Method:

- Receive the transaction identification information from checkPaymentStatusDto.

- Call the VietQR gateway to synchronize and verify the current transaction status.
- Process the returned transaction data and update the internal system with the latest payment state.
- Return a `PaymentStatusDto` representing the synchronized transaction status.

Design for class "VietqrApiClient"



Description: API client class responsible for communicating with the external VietQR system. It handles authentication, fetching access tokens, requesting QR code generation, and querying transaction statuses directly from the VietQR Sandbox.

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	clientId	string		The Client ID provided by VietQR for authentication.
2	apiKey	string		The API Key provided by VietQR for authentication.
3	accessToken	string		The temporary access token retrieved from VietQR to authorize subsequent API calls.
4	baseUrl	string	"https://api.vietqr.vn/"	The base URL for the VietQR Sandbox API.

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	getAccessToken	string	Authenticate with VietQR using clientId and apiKey to retrieve a new access token.

#	Name	Return type	Description (purpose)
2	generateQRCode	VietqrGenerateResponseDto	Send a request to the VietQR API to generate a new QR code for an order.
3	getTransactionStatus	VietqrTransactionStatusResponseDto	Query the VietQR API to synchronize and check the current status of a specific transaction.

Detailed Operation Design

1. Operation : getAccessToken

Exception:

- PaymentException if the API call fails, network connection is lost, or authentication is rejected by VietQR due to invalid credentials.

Method:

- Construct the authentication request using the clientId and apiKey attributes.
- Call the VietQR "Get Access Token" endpoint (/vi/api-vietqr-callback/goi-api-get-token).
- Extract the access token from the response.
- Update the internal accessToken attribute and return the token string for use in subsequent requests.

2. Operation: generateQRCode

Parameter:

- request: VietqrGenerateRequestDto The formatted request payload containing order and account details necessary to create the QR.
- token: string The active access token retrieved from getAccessToken.

Exception:

- - PaymentException if the API call fails, the token is expired/invalid, or the request payload is rejected by VietQR.

Method:

- Construct the HTTP request with the token in the Authorization header.
- Attach the request payload to the body.
- Call the VietQR "Generate QR Code" endpoint (/vi/api-vietqr-callback/goi-api-generate-vietqr-code).
- Parse the raw JSON response from VietQR.
- Map and return the data as a VietqrGenerateResponseDto.

3. Operation 3: getTransactionStatus**Parameter:**

- checkPaymentStatusDto: CheckPaymentStatusDto DTO containing identifiers (like transactionId or orderReference) to locate the specific transaction.
- token: string The active access token retrieved from getAccessToken.

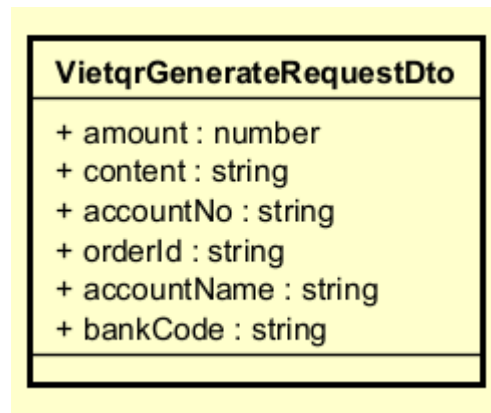
Exception:

- PaymentException if the API call fails, the token is expired/invalid, or the transaction cannot be found on the VietQR system.

Method:

- Extract the necessary transaction identifier from checkPaymentStatusDto.
- Construct the HTTP request using the token for authorization.
- Call the VietQR "Transaction Sync" endpoint (/vi/api-vietqr-callback/api-transaction-sync) to retrieve the current state.
- Parse the raw JSON response from VietQR.
- Map and return the status details as a VietqrTransactionStatusResponseDto.

Design for class "VietqrGenerateRequestDto"



Description: Data Transfer Object (DTO) containing the necessary payload information sent to the VietQR API to request the generation of a new QR code for an order.

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	amount	number		The total payment amount required for the order.
2	content	string		The transaction description or transfer message to be embedded in the QR code.
3	accountNo	string		The merchant's receiving bank account number.
4	orderId	string		The internal system's unique identifier for the order.
5	accountName	string		The registered name of the merchant's bank account.

#	Name	Data type	Default value	Description
6	bankCode	string		The BIN or code of the merchant's receiving bank.

Design for class "VietqrGenerateResponseDto"

VietqrGenerateResponseDto
+ qrCode : string + qrLink : string + bankCode : string + bankName : string + bankAccount : string + transactionId : string + amount : number + content : string + existing : boolean + userBankName : string + transactionRefId : string + imgId : string + terminalCode : string + subTerminalCode : string + serviceCode : string + orderId : string + additionalData : string + vaAccount : string

Description: Data Transfer Object (DTO) representing the comprehensive payload returned by the VietQR system upon successfully generating a QR code.

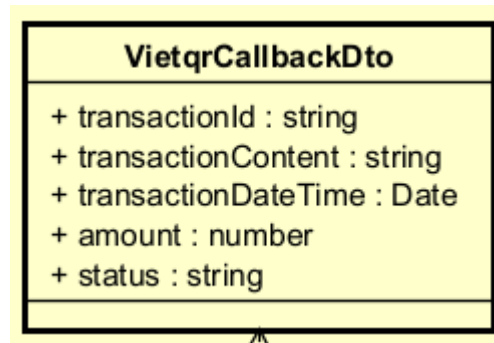
Table 1. Attribute design

#	Name	Data type	Default value	Description
1	qrCode	string		The raw text or Base64 string representation of the generated QR code.
2	qrLink	string		A direct URL link to an image of the generated QR code.
3	bankCode	string		The code of the receiving bank.

#	Name	Data type	Default value	Description
4	bankName	string		The name of the receiving bank.
5	bankAccount	string		The receiving bank account number.
6	transactionId	string		VietQR's unique identifier for this generation transaction.
7	amount	number		The payment amount encoded in the QR.
8	content	string		The transaction description encoded in the QR.
9	existing	boolean		Indicates if a matching QR code already existed.
10	userBankName	string		The bank name associated with the user/merchant profile.
11	transactionRefId	string		An external reference ID for the transaction.
12	imgId	string		Identifier for the generated QR image asset.
13	terminalCode	string		Code identifying the payment terminal (if applicable).
14	subTerminalCode	string		Code identifying the sub-terminal.

#	Name	Data type	Default value	Description
15	serviceCode	string		The specific service code used for generation.
16	orderId	string		The internal system's order identifier linked to this QR.
17	additionalData	string		Any extra metadata or payload data returned by the API.
18	vaAccount	string		The Virtual Account number assigned to this transaction, if applicable.

Design for class "VietqrCallbackDto"



Description: Data Transfer Object (DTO) defining the structure of the incoming webhook/callback payload sent by the VietQR system when a transaction status changes (e.g., successful payment).

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	transactionId	string		The unique identifier of the payment transaction on the VietQR system.
2	transactionContent	string		The actual transfer message or description sent by the customer during payment.
3	transactionDateTime	Date		The timestamp of when the payment transaction occurred.
4	amount	number		The actual monetary amount successfully transferred by the customer.
5	status	string		The final status of the transaction (e.g., "SUCCESS", "FAILED").

Design for class "VietqrAccessTokenResponseDto"

Description: Description: VietqrAccessTokenResponseDto is a data transfer object that stores the access token information returned by the VietQR gateway after successful authentication. It includes the access token, token type, and token expiration duration, which are used for subsequent authorized API operations.

VietqrAccessTokenResponseDto
+ accessToken : string + tokenType : string + expiresIn : number

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	accessToken	string		Stores the access token returned by the VietQR gateway after successful authentication. This token is used to authorize subsequent API requests.
2	tokenType	string		Specifies the type of the returned access token, typically used together with the token value in the authorization header.
3	expiresIn	number		Indicates the validity period of the access token, usually expressed in seconds before the token expires.

Design for class "VietqrAccessTokenRequestDto"



Description: VietqrAccessTokenRequestDto is a data transfer object used to carry the authentication credentials required to request an access token from the VietQR gateway. It contains the client identifier and API key, which are used to authenticate the system before calling subsequent VietQR APIs.

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	clientId	string		The client identifier issued by the VietQR gateway, used to identify the requesting system during authentication.
2	apiKey	string		The API key provided by the VietQR gateway, used together with the client identifier to authenticate and obtain an access token.